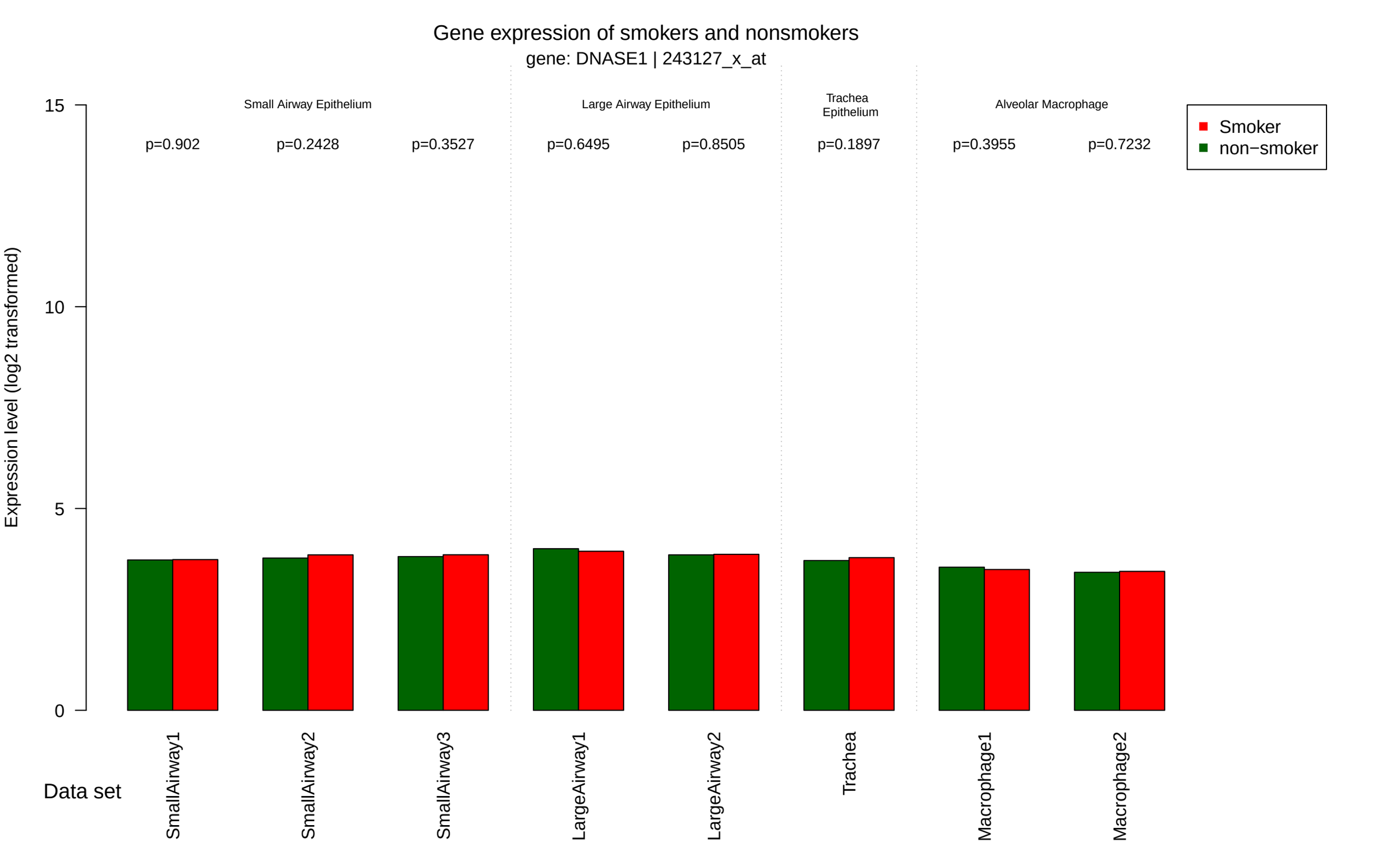
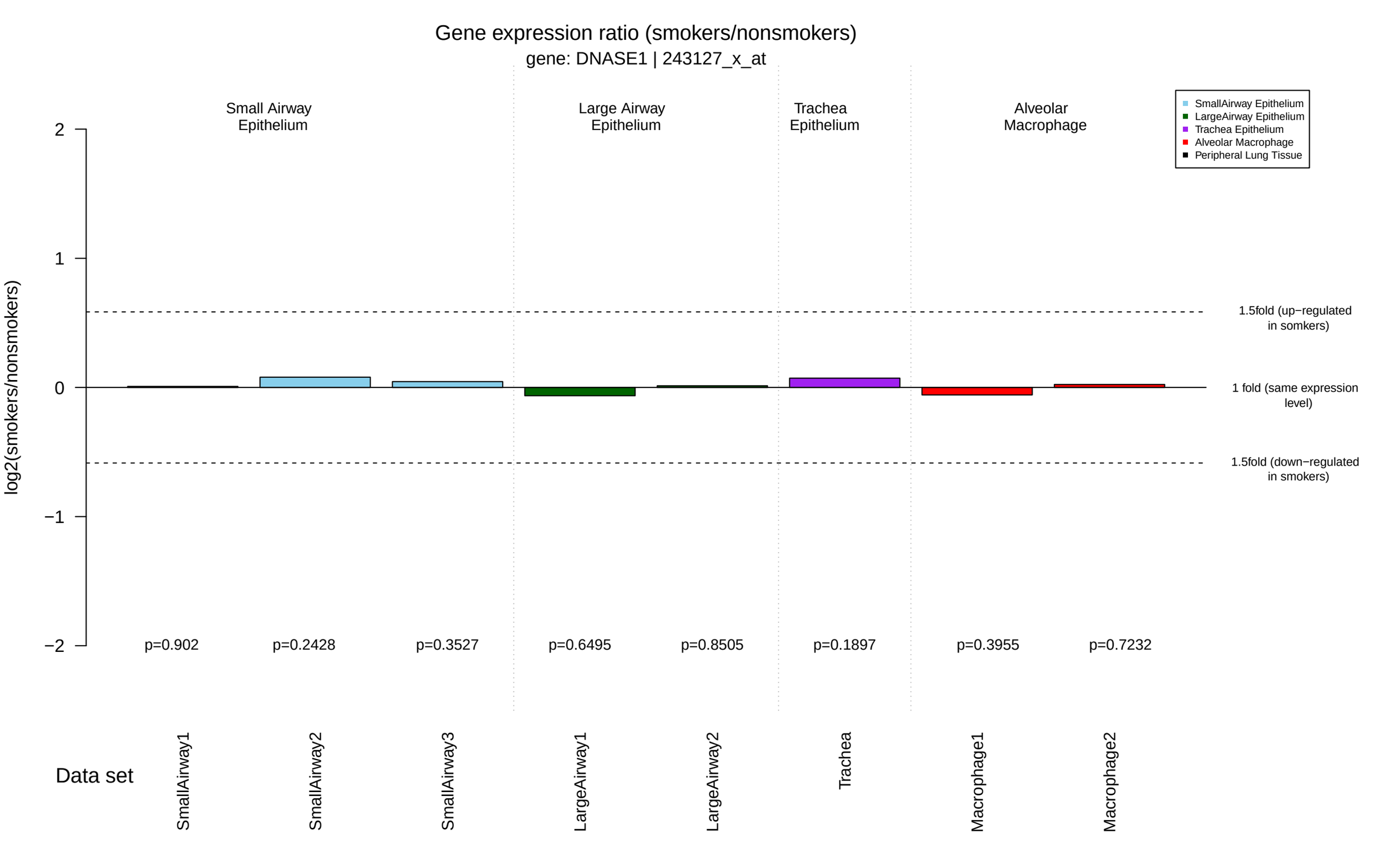
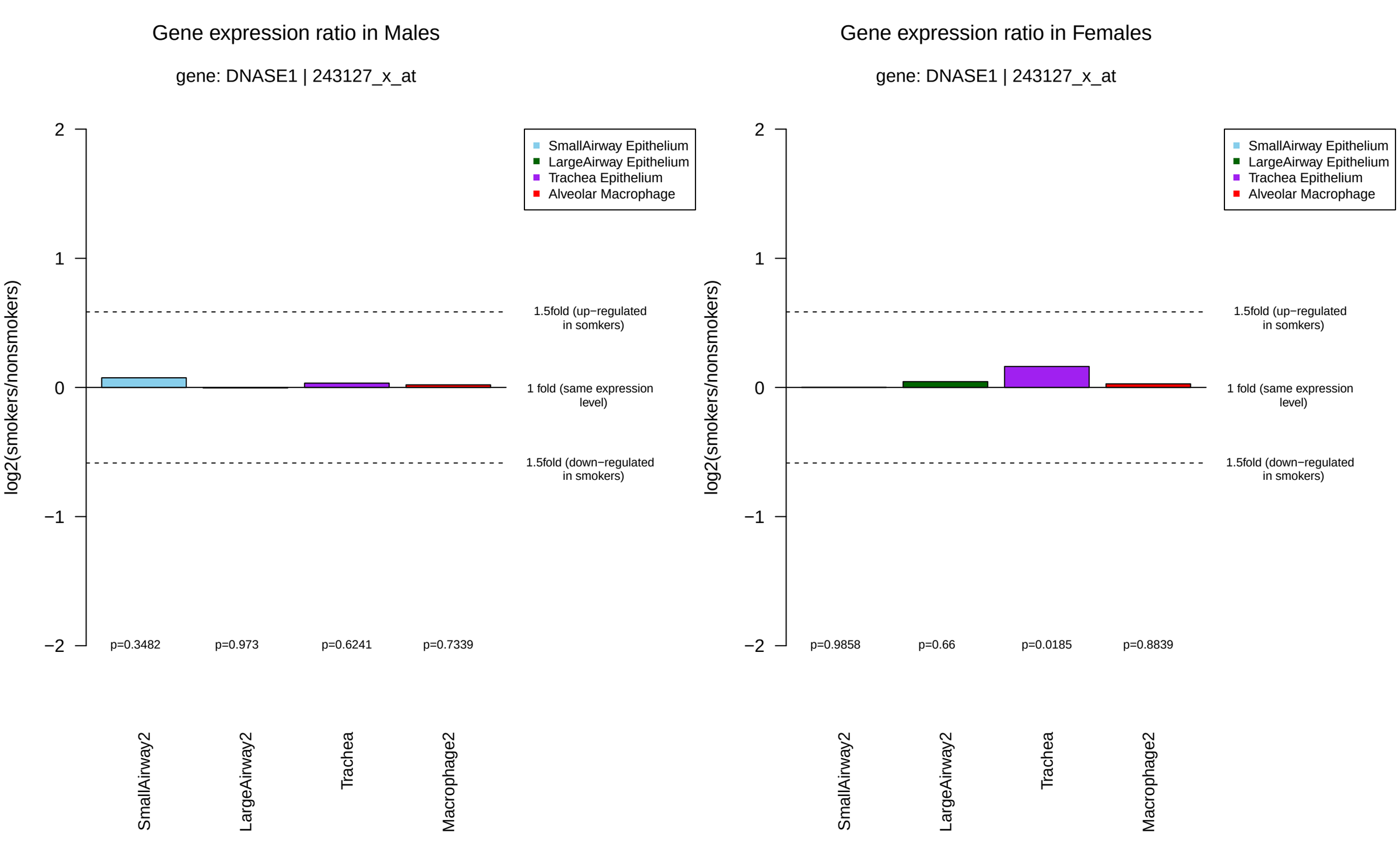
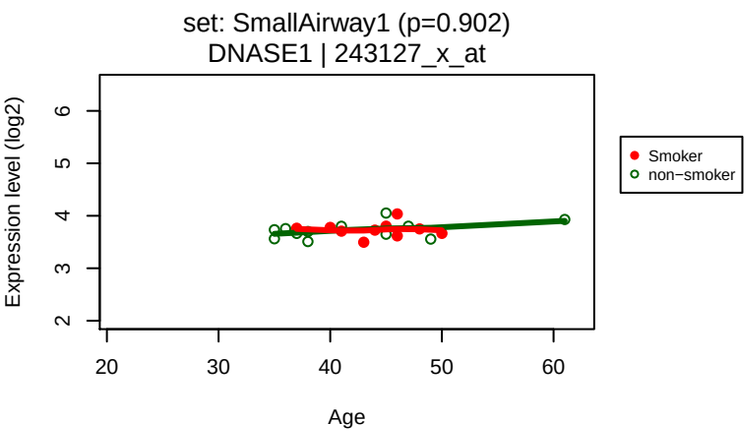
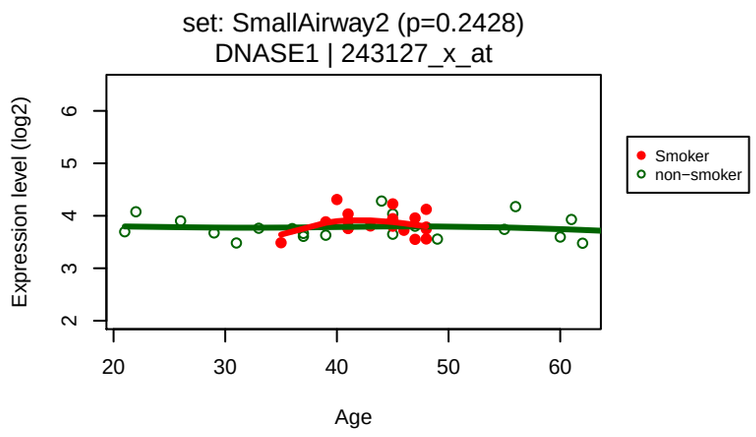
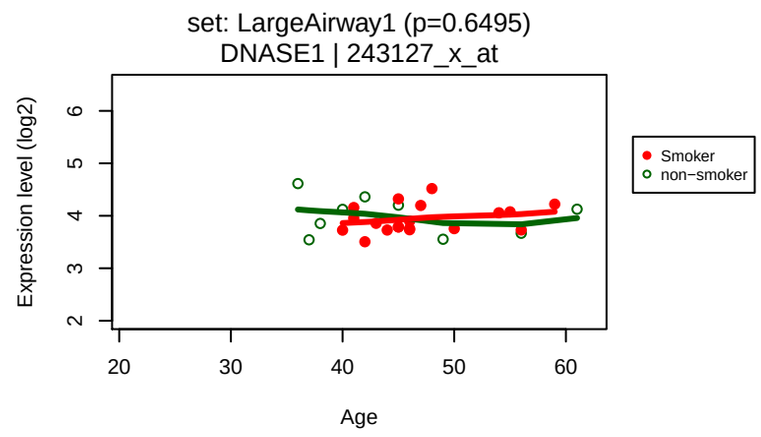

Gene expression of smokers and nonsmokers
gene: DNASE1 | 243127_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.902
p=0.2428
p=0.3527
p=0.6495
p=0.8505
p=0.1897
p=0.3955
p=0.7232
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

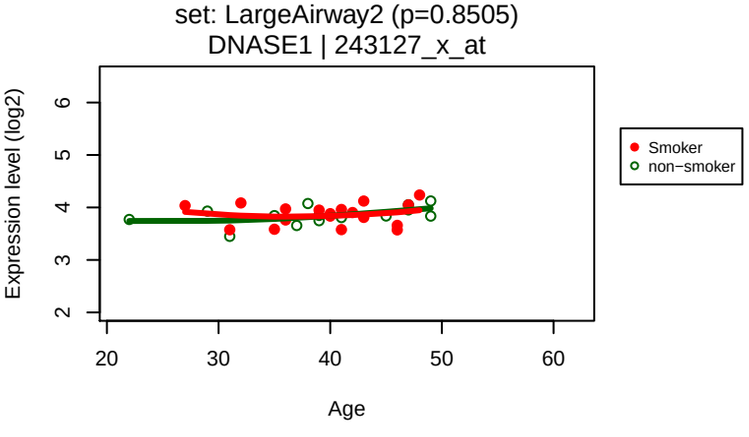

Gene expression ratio (smokers/nonsmokers)
gene: DNASE1 | 243127_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.902
p=0.2428
p=0.3527
p=0.6495
p=0.8505
p=0.1897
p=0.3955
p=0.7232
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

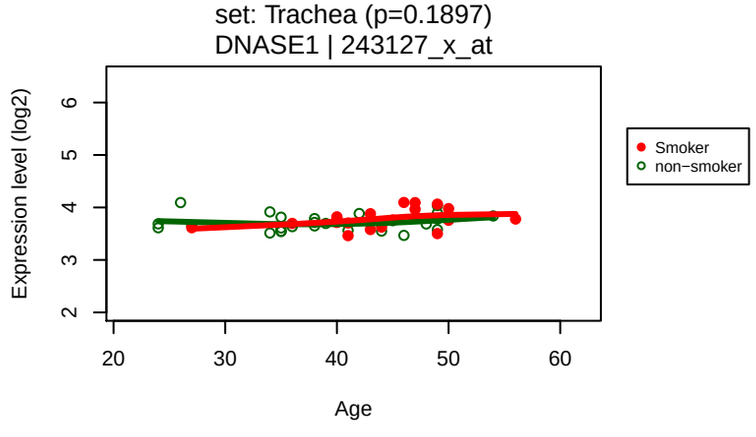

Gene expression ratio in Males
gene: DNASE1 | 243127_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3482
p=0.973
p=0.6241
p=0.7339
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNASE1 | 243127_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9858
p=0.66
p=0.0185
p=0.8839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

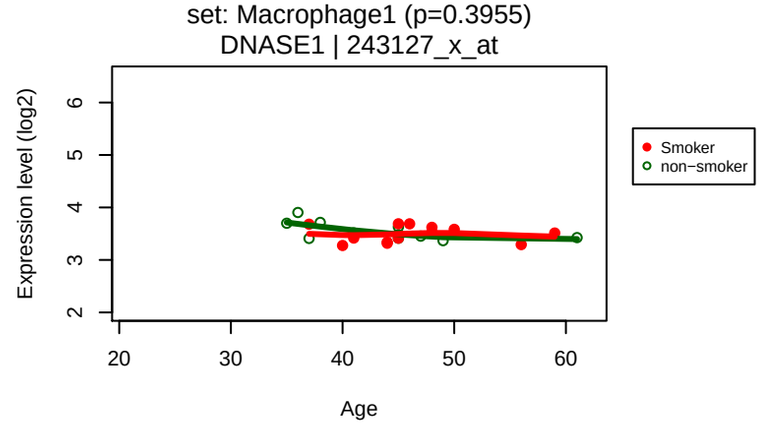

set: SmallAirway1 (p=0.902)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker

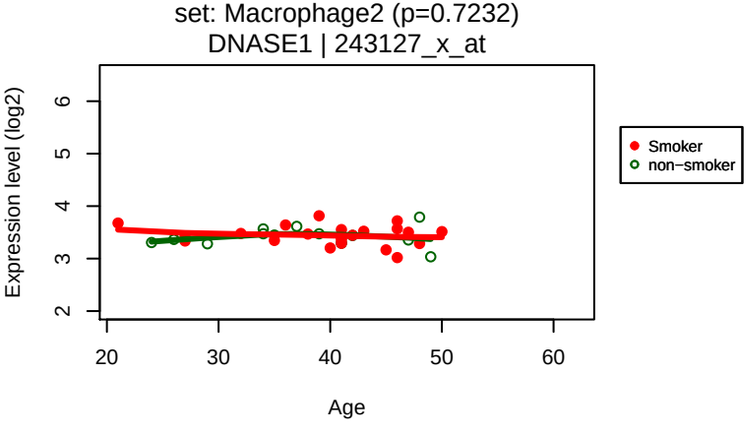

set: SmallAirway2 (p=0.2428)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker

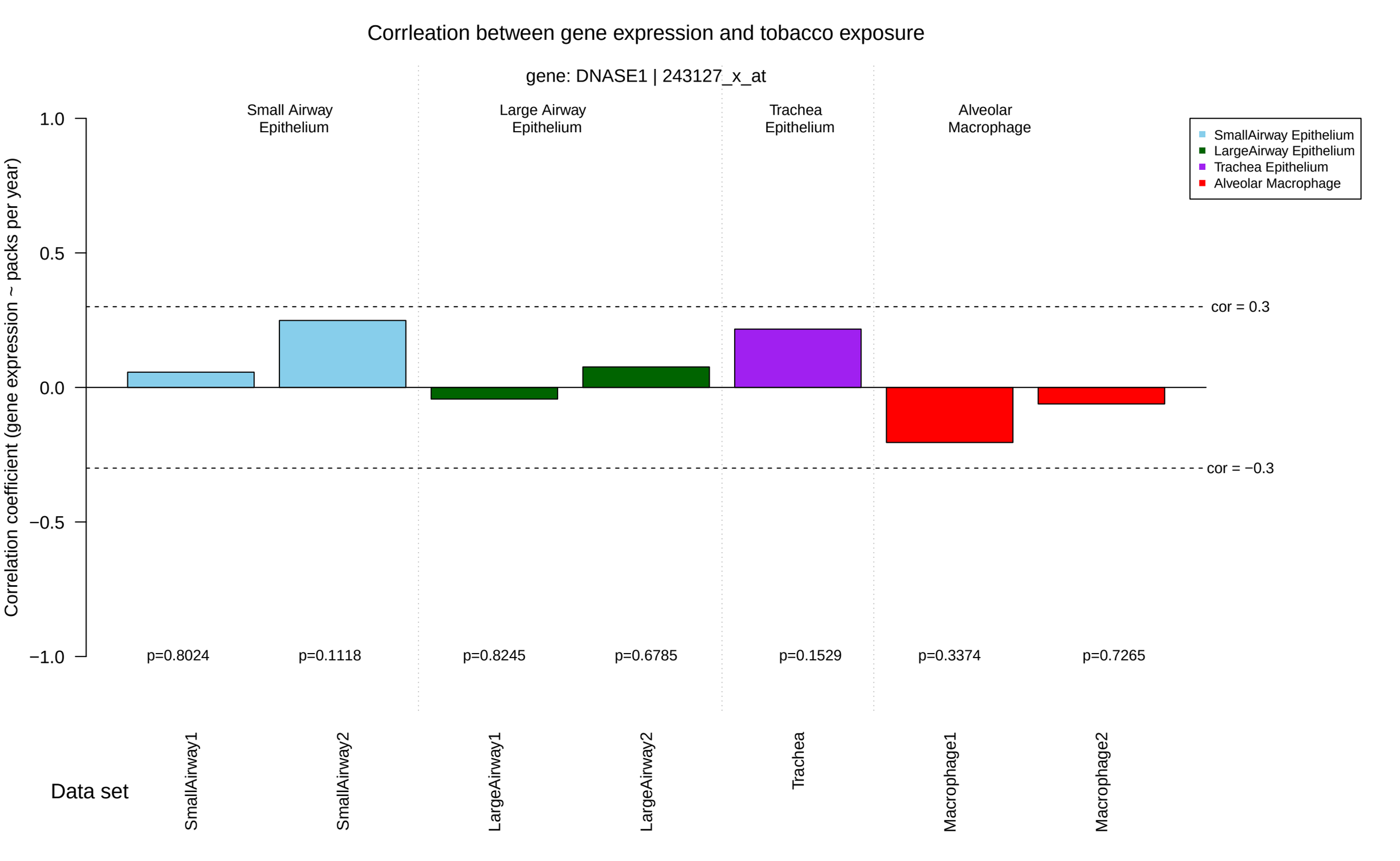

set: LargeAirway1 (p=0.6495)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8505)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1897)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3955)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7232)
DNASE1 | 243127_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNASE1 | 243127_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8024
p=0.1118
p=0.8245
p=0.6785
p=0.1529
p=0.3374
p=0.7265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2